MAAI

Step-by-Step Hackathon Build Roadmap

From an empty project folder to a deployed, end-to-end voice-first maternal mental-health screening prototype.

Target submission date: 31 October 2026

Core stack: Next.js + TypeScript + Tailwind · FastAPI · Supabase · Bolna · Vercel

Build principle: get one complete path working first — consent → phone call → webhook → backend → database → visible result — then expand into scoring, escalation and the dashboard.

Based on the MAAI one-pager and the implementation roadmap discussed in chat.

WHAT YOU WILL BUILD

Layer	Purpose
Mother / ASHA interface	Register a mother, record consent and set language/contact preferences.
Voice orchestration	Start an outbound call and manage the conversation through Bolna.
Backend	Validate events and answers, calculate approved screening scores, and apply safety rules.
Data layer	Persist consent, sessions, answers, results, alerts and follow-up status.
Clinician dashboard	Review screening history, acknowledge alerts and record follow-up actions.

1. Recommended Infrastructure

The following is the recommended implementation architecture. The source one-pager names Bolna, speech providers, FastAPI, Supabase, Next.js and Vercel; scheduler details, event handling and access-control boundaries are implementation recommendations.

```
Mother / ASHA worker
|
v
Next.js + Tailwind (registration + clinician dashboard)
| HTTPS API
v
Supabase PostgreSQL + Auth + Row-Level Security
^ |
| v
FastAPI <--- Bolna <--- Scheduler / eligible call trigger
| |
| +--> Speech recognition / LLM / text-to-speech
|
+--> Validate answers --> Deterministic scoring --> Safety rules
|
v
Alert + human follow-up workflow
|
v
Clinician dashboard
```

Technology responsibilities

Component	Responsibility
Next.js + Tailwind	Registration interface and clinician dashboard.
Supabase	PostgreSQL, authentication and row-level security.
Bolna	Outbound calls and voice-agent orchestration.
Speech provider	Speech recognition and text-to-speech, based on supported integrations.
LLM	Conversation wording, turn management and clarification — not clinical scoring.
FastAPI	Validate webhook events, store answers, calculate score and apply safety rules.
Scheduler	Initiate consented check-ins at permitted times.
Vercel + Python host	Host frontend and backend separately.

Important integration rule

Confirm which speech providers and phone providers are supported by the current Bolna integration. Do not assume providers can be swapped without configuration. Inspect real webhook payloads before implementing event parsing.

2. Milestone 1 — Foundation and Setup

Target: 24 October

Create the repository and start the frontend and backend.

```
maai/
■■■ frontend/ # Next.js app
■■■ backend/
■ ■■■ app/
■ ■ ■■■ main.py
■ ■ ■■■ routes/
■ ■ ■■■ schemas/
■ ■ ■■■ services/
```

```
■ ■ ■■■ scoring/
■ ■ ■■■ safety/
■ ■ ■■■ database/
■ ■■■ tests/
■ ■■■ requirements.txt
■■■ supabase/migrations/
■■■ docs/
■■■ README.md
```

Create the frontend

- Run: npx create-next-app@latest frontend --typescript --tailwind --app
- Create a Python virtual environment in backend/ and activate it.
- Install initial dependencies: fastapi, uvicorn[standard], pydantic-settings, supabase and httpx.

Minimal backend health check

```
from fastapi import FastAPI

app = FastAPI(title="MAAI API")

@app.get("/health")
def health():
return {"status": "ok", "service": "maai-api"}

# Run from backend/: uvicorn app.main:app --reload
```

Acceptance criteria

- Frontend runs locally; FastAPI /health returns OK.
- Supabase project and GitHub repository exist.
- Secrets are stored in environment variables and never committed.

3. Milestone 2 — Registration and Consent

Target: 25 October

Build the registration form with name, phone, preferred language, registration source, assigned worker, consent status and permitted contact window. Use fictional records or consenting volunteers.

Initial tables

Table	Purpose / example fields
mothers	id, name, phone, preferred_language, created_by
consents	mother_id, consent_type, status, granted_at, withdrawn_at
screening_sessions	id, mother_id, provider_call_id, status, started_at, completed_at

Suggested endpoints

```
POST /api/mothers
GET /api/mothers
GET /api/mothers/{mother_id}
POST /api/consents
POST /api/consents/{consent_id}/withdraw
```

Acceptance criteria

- Mother registration persists after refresh.
- Consent is visible and can be withdrawn.
- Backend checks consent immediately before dispatching a call.

- Unauthorized users cannot access records outside their permissions.

4. Milestone 3 — Voice Agent and Telephony

Target: 25–26 October

- Create a Bolna account and check available buildathon credits and integrations.
- Configure an outbound agent, supported phone provider and language/speech settings.
- Make a basic test call before adding the full questionnaire.
- Configure the supported webhook to reach the backend over public HTTPS.

Conversation state machine

```
Call connected -> Introduce MAAI -> Confirm recipient
-> Explain purpose and request consent
-> Declines: record outcome and end call
-> Agrees: ask approved question
-> Unclear answer: clarify / retry
-> Valid answer: persist structured response
-> More questions: continue
-> Complete: backend scoring and safety workflow
```

Webhook handling

- Verify incoming events using the provider-supported authentication mechanism.
- Identify the call and screening session; validate the payload schema.
- Persist the event and update session state.
- Make event processing idempotent so replayed events do not duplicate answers.
- Inspect actual provider payloads; do not assume event field names.

Acceptance criteria

- A real test call connects and the introduction/consent flow works.
- Call status and structured answers reach FastAPI.
- Dropped or failed calls are recorded as incomplete.

5. Milestone 4 — Screening, Scoring and Safety

Target: 26–27 October

- Use an approved EPDS questionnaire and language version with stable question IDs and valid answer codes.
- Map captured responses to approved codes; ask for clarification instead of guessing.
- Implement scoring in backend/app/scoring/epds.py, including completeness checks and approved scoring rules.
- Store the scoring protocol version with the result.
- Implement reviewed escalation rules in backend/app/safety/rules.py.
- Separate screening score, risk category, urgent safety signals and human escalation status.

Suggested alert flow

```
Safety rule matches -> Create alert -> Assign reviewer
-> Attempt notification -> Track delivery
-> Human acknowledges -> Record next action
-> Close only according to the approved workflow
```

Tests and acceptance criteria

- Scoring matches approved reference cases.
- Incomplete questionnaires do not receive completed scores.

- Urgent synthetic scenarios trigger the reviewed response workflow independently of total score.
- Duplicate events do not duplicate answers or alerts.
- Failed notifications remain visible with a defined retry/fallback procedure.

6. Milestone 5 — Clinician Dashboard

Target: 27–28 October

Screen	What it should show
Overview	Scheduled calls, completed and incomplete screenings, alerts awaiting review and follow-ups due.
Mother profile	Consent history, call history, screening history and follow-up status.
Alert queue	Reason, urgency, assigned reviewer, acknowledgement status and next action.
Follow-up workspace	Owner, action notes, due date, status and completion record.

- Populate metrics from persisted records, not hardcoded values.
- Use authenticated API requests and enforce authorization server-side.
- Apply Supabase Row Level Security and restrict transcript access.
- Do not expose sensitive data in public bundles, logs or analytics.

Acceptance criteria

- Dashboard reflects backend records.
- Clinician can acknowledge an alert and record follow-up.
- Incomplete and completed screenings are clearly distinguished.

7. Milestone 6 — End-to-End Integration and Testing

Target: 29 October

Scenario	Expected behavior
Normal screening	Answers are captured, validated, scored and displayed.
Consent withdrawn	No subsequent scheduled call is initiated.
Incomplete call	Session stays incomplete; no completed score is assigned.
Unclear answer	Agent clarifies rather than silently guessing.
Urgent disclosure	Reviewed safety flow interrupts normal path and escalates.
Duplicate webhook	No duplicate answers, sessions or alerts.
Notification failure	Failure is visible and retry/fallback runs.
Unauthorized access	User cannot access records outside their permissions.

Trace every test across the call record, webhook, FastAPI logs, Supabase session/answers, result/alert and clinician dashboard. Use correlation IDs and timestamps. Avoid logging raw phone numbers, full transcripts or unnecessary health data.

Acceptance criteria

- At least one normal and one escalation scenario work end to end.
- Failure scenarios behave predictably.

- Backend restarts do not lose persisted screening state.

8. Milestone 7 — Deployment

Target: 30 October

Component	Deployment target
Frontend	Vercel
Backend	Python-capable hosting service
Database	Supabase
Voice agent	Bolna hosted service
Scheduler	Backend scheduler or managed scheduled job
Notifications	Configured notification provider

- Set production environment variables and verify HTTPS.
- Configure webhook URL and authentication in Bolna.
- Verify CORS, database policies, timeouts, provider limits and error handling.
- Test the complete deployed application.
- Prepare a recorded backup demo and label simulated events honestly.

9. Milestone 8 — Demo and Submission

Target: 31 October

Time	Demo segment
0:00–0:30	Problem: access barriers to timely maternal mental-health screening.
0:30–1:30	Register a fictional mother, show consent and demonstrate a test voice call.
1:30–2:30	Show backend result, deterministic scoring, synthetic alert and clinician follow-up.
2:30–3:00	Demonstrate consent withdrawal and explain human handoff and safety.

Pitch deck outline

- Problem and target users.
- Solution and live demo.
- Architecture and why scoring is deterministic.
- Differentiation: Indian-language phone access and human referral workflow.
- Safety boundaries, consent and privacy.
- Distribution through ASHA workers/hospitals and next steps.

10. Implementation Reference

Suggested FastAPI endpoints

```
POST /api/mothers
POST /api/consents
POST /api/screenings
POST /api/screenings/{session_id}/answers
POST /api/webhooks/bolna
GET /api/screenings/{session_id}
GET /api/alerts
POST /api/alerts/{alert_id}/acknowledge
POST /api/alerts/{alert_id}/followups
POST /api/consents/{consent_id}/withdraw
```

These are proposed endpoint names, not existing provider endpoints. Implement the Bolna webhook to match the current provider documentation and actual payloads.

Recommended data tables

Table	Core fields
mothers	id, name, phone, preferred_language, created_by
consents	mother_id, consent_type, status, granted_at, withdrawn_at
screening_sessions	id, mother_id, provider_call_id, status, started_at, completed_at
screening_answers	session_id, question_code, answer_code, recorded_at
screening_results	session_id, score, protocol_version, review_status
safety_alerts	session_id, severity, reason_code, status, acknowledged_at
followups	alert_id, assigned_to, due_at, outcome, completed_at
audit_logs	actor_id, action, record_type, record_id, created_at

Three database rules

- Preserve validated answer codes and scoring protocol version for auditability.
- Keep incomplete calls distinguishable from completed screenings.
- Enforce permissions server-side and with Row Level Security.

Definition of done

- Mother registration and consent work.
- Outbound test call and structured answer capture work.
- FastAPI validates and persists answers.
- Approved scoring rules run deterministically.
- Synthetic safety scenario creates an alert.
- Clinician can acknowledge and record follow-up.
- Consent withdrawal prevents future scheduled calls.
- Deployed application works end to end.
- Backup demo and pitch deck are ready.

11. Safety and Scope Boundaries

MAAI is a screening, support and referral prototype — not diagnosis or treatment. EPDS is a screening instrument. Use a clinically approved questionnaire, wording, scoring rubric and escalation protocol. Urgent disclosures must follow a reviewed response path and must not wait for the total score.

- Use synthetic or consenting volunteer data for the hackathon demo.
- A notification alone is not emergency care; define human response and failure procedures before real-patient use.
- Before production use, complete appropriate clinical review, privacy safeguards, access controls, retention policies and reliable human escalation arrangements.
- Do not infer depression from voice tone or represent wellbeing trends as diagnosis.

Recommended build order

Consent → Bolna test call → Webhook → FastAPI → Supabase → Visible result. Then add approved scoring and safety workflows, build the clinician dashboard, test end to end, deploy and rehearse the demo.

Planning note: milestone dates are targets derived from the submission date in the source one-pager; actual progress depends on provider access and integration complexity.